



RISC-V External Debug Security Specification

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Preamble



This document is in the *Development state*

Expect potential changes. This draft specification is likely to evolve before it is accepted as a standard. Implementations based on this draft may not conform to the future standard.

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Chapter 1. Introduction

Debugging and tracing are essential for developers to identify and rectify software and hardware issues, optimize performance, and ensure robust system functionality. The debugging and tracing extensions in the RISC-V ecosystem play a pivotal role in enabling these capabilities, allowing developers to monitor and control the execution of programs during the development, testing, and production phases. However, the current RISC-V Debug specification grants the external debugger the highest privilege in the system, regardless of the privilege level at which the target system is running, and tracing is also allowed in all privilege modes. This leads to privilege escalation and information leakage issues when multiple actors are present.

This specification defines [Debug Module Security Extension \(non-ISA extension\)](#) and [Sdsec \(ISA extension\)](#) to address the above security issues in the current *The RISC-V Debug Specification* [1] and trace specifications [2] [3].

At a high level, the Debug Module Security Extension provides a platform-level control state, `psecdngen`, to enable or bypass the security constraints. The Sdsec ISA extension provides per-hart debug and trace control states, including platform-controlled `mdngen` and `mtrcen` states for M-mode and lower-privilege control states in `mdtcfg`. Together, these control states determine whether external debug or trace is allowed at the hart's current privilege mode.

A summary of the changes introduced by *The RISC-V External Debug Security Specification* follows.

- **Per-Hart Debug Control:** Introduce per-hart states to control whether external debug is allowed for each privilege mode.
- **Per-Hart Trace Control:** Introduce per-hart states to control whether tracing is allowed for each privilege mode.
- **Platform debug security enable:** Add a platform state to enable the security constraints or bypass them for backward compatibility.
- **Debug Mode entry:** An external debugger can only halt the hart and enter Debug Mode when external debug is allowed in the current privilege mode.
- **Memory Access:** Memory access from a hart's point of view, using the Program Buffer or an Abstract Command, must be checked by the hart's memory protection mechanisms as if the hart is running at [debug access privilege level](#); memory access from the Debug Module using System Bus Access (SBA) must be checked by a system memory protection mechanism, such as IOPMP or RISC-V Worlds.
- **Register Access:** Register access using the Program Buffer or an Abstract Command works as if the hart is running at [debug access privilege level](#) instead of the M-mode privilege level. The debug CSRs (`dcsr` and `dpc`) are shadowed in lower privilege modes.

1.1. Control Hierarchy

Operationally, the platform-level control state selects between backward-compatible behavior and the security-constrained behavior defined by this specification. When the constraints are enabled, each hart evaluates its debug and trace control states for its current privilege mode. Debug operations are permitted only when the corresponding debug control state allows access at that mode; otherwise the request is rejected or remains pending. Trace output is similarly enabled only for modes allowed by the trace control states and is inhibited otherwise.

1.1.1. External Debug Security Control

The debug security policy for a hart is enforced through a hierarchical control model using the platform-

controlled **mdbgen** state for M-mode and optional lower-privilege control states (**SEDBGEN**, **VSEDBGEN**, **UEDBGEN**, **VUEDBGEN**) in the **mdtcfg** CSR. The **mdbgen** state can be implemented as an input signal to the hart and managed through platform-specific mechanisms such as Debug Module registers, a Root of Trust (RoT) handshake, or lifecycle fuses. The **psecdngen** platform control state enables the debug security policy at the platform level; when **psecdngen** is 0, the platform provides unrestricted debug access for all privilege modes.

The debug security control logic validates all debug requests and triggers whose **ACTION**=1 against the hart's current privilege level and the configured debug security control states. The validation procedure involves two steps: (1) checking **psecdngen** to determine whether the debug security policy is enabled and (2) checking the hierarchical control states to determine whether external debug is allowed at the hart's current privilege level. Debug requests that fail validation are either dropped or kept pending until the hart transitions to a debug-allowed privilege mode. [External Debug Security Extensions](#) defines the detailed behavior.

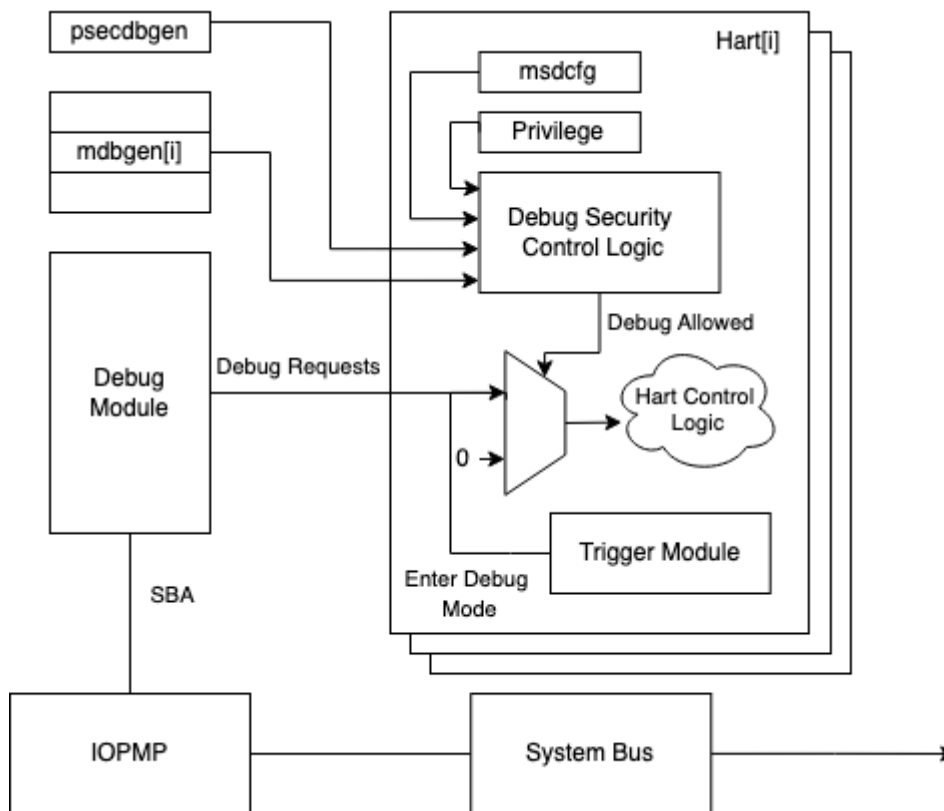


Figure 1. The debug security control for one hart

The hierarchical control applies as follows for a single hart:

- When **psecdngen**=0, external debug is allowed for all privilege modes.
- When **psecdngen**=1 and **mdbgen**=1, external debug is allowed for all privilege modes.
- When **psecdngen**=1, **mdbgen**=0, and **SEDBGEN**=1 (if implemented), external debug is allowed for S/HS-mode and lower privilege modes.
- When **psecdngen**=1, **mdbgen**=0, **SEDBGEN**=0, and **VSEDBGEN**=1 (if implemented), external debug is allowed for VS-mode and VU-mode.
- When **psecdngen**=1, **mdbgen**=0, **SEDBGEN**=0, and **UEDBGEN**=1 (if implemented), external debug is allowed for U-mode. **VSEDBGEN** and **VUEDBGEN** do not affect U-mode debug permission.
- When **psecdngen**=1, **mdbgen**=0, **SEDBGEN**=0, **VSEDBGEN**=0, and **VUEDBGEN**=1 (if implemented), external debug is allowed for VU-mode. **UEDBGEN** does not affect VU-mode debug permission.
- External debug is disallowed in a privilege mode when none of the control states applicable to that

mode allow debug.

The debug security control logic must ensure that debug requests are validated and executed at the same privilege level and with the same debug security control states to prevent time-of-check to time-of-use (TOCTOU) vulnerabilities. Failing to do so may allow debug requests to bypass security controls.



Self-hosted debugging is also commonly used for application-level debugging, providing a pure software-based solution that does not require an external debugger. It is orthogonal to external debugging and is not affected by the debug security controls.

1.1.2. Trace Security Control

The trace security policy for a hart is enforced through a hierarchical control model using the platform-controlled `mtrcen` state for M-mode and optional lower-privilege control states (`SETRCEN`, `VSETRCEN`, `UETRCEN`, `VUETRCEN`) in the `mdtcfg` CSR. The `mtrcen` state can be implemented as an input signal to the hart and managed through platform-specific mechanisms such as a Root of Trust (RoT) handshake or lifecycle fuses.

The security control logic determines whether trace is allowed based on the hart's current privilege level and the configured trace security control states. The `sec_inhibit` sideband signal to the trace encoder is asserted when trace is not allowed; equivalently, `sec_inhibit` is the inverse of the trace-allowed decision shown in the reference control logic. [Trace Security Extensions](#) defines the detailed behavior.

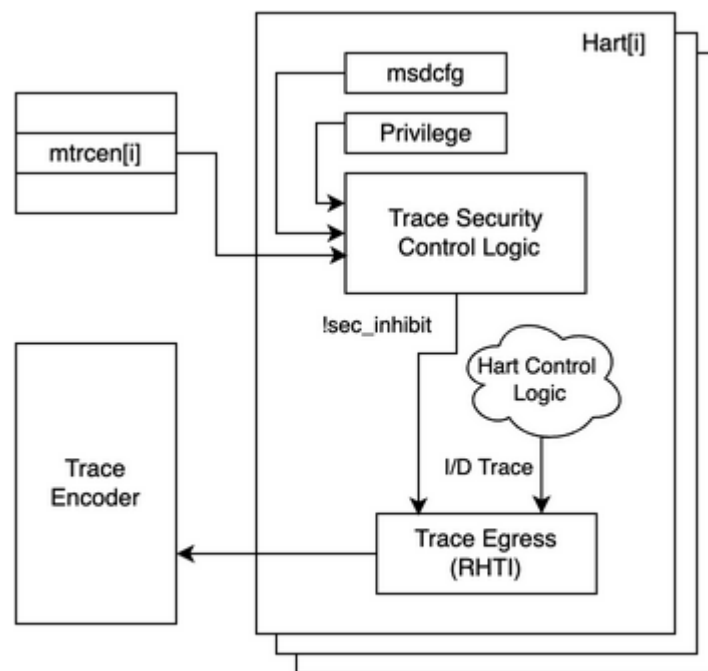


Figure 2. The trace security control for one hart

The hierarchical control applies as follows for a single hart:

- When `mtrcen`=1, trace is allowed for all privilege modes.
- When `mtrcen`=0 and `SETRCEN`=1 (if implemented), trace is allowed for S/HS-mode and lower privilege modes; trace is inhibited for M-mode.
- When `mtrcen`=0, `SETRCEN`=0, and `VSETRCEN`=1 (if implemented), trace is allowed for VS-mode and VU-mode; trace is inhibited for M-mode and S/HS-mode.

- When `mtrcen=0`, `setrcen=0`, and `uetrcen=1` (if implemented), trace is allowed for U-mode. `vsetrcen` and `vuetracen` do not affect U-mode trace permission.
- When `mtrcen=0`, `setrcen=0`, `vsetrcen=0`, and `vuetracen=1` (if implemented), trace is allowed for VU-mode. `uetrcen` does not affect VU-mode trace permission.
- When all trace control states are 0, trace is completely inhibited for all privilege modes.

1.2. Terminology

Abstract command	A high-level Debug Module operation used to interact with and control harts
Debug Access Privilege	The privilege level with which an Abstract Command or instruction in the Program Buffer accesses hardware resources
Debug Mode	An additional privilege mode to support off-chip debugging
Hart	A RISC-V hardware thread
IOPMP	Input-Output Physical Memory Protection unit
M-mode	The highest privileged mode in the RISC-V privilege model
PMA	Physical Memory Attributes
PMP	Physical Memory Protection unit
Program buffer	A mechanism that allows the Debug Module to execute arbitrary instructions on a hart
RISC-V Worlds	A system-level isolation mechanism that constrains physical-address access by harts and other bus initiators
Trace encoder	A piece of hardware that takes in instruction execution information from a RISC-V hart and transforms it into trace packets

Chapter 2. External Debug Security Threat Model

Modern SoC development involves several different actors who may not trust each other, resulting in the need to isolate actors' assets during the development and debugging phases. The current RISC-V Debug specification [1] grants external debuggers the highest privilege in the system, regardless of the privilege level at which the target system is running. This leads to privilege escalation issues when multiple actors are present.

For example, the owner of an SoC, who needs to debug their firmware, may be able to use the external debugger to bypass PMP (including PMP lock `pmpecfg.L=1`) and MMU protections, attacking Boot ROM or other SoC creator's assets.

Additionally, the RISC-V privilege architecture supports multiple software entities at different privilege levels that may not trust each other. These entities may be isolated from each other by mechanisms such as PMP/IOPMP or MMU, and may need different debug and trace policies. Since the external debugger is granted the highest privilege in the system, a malicious entity at a lower privilege level could compromise higher privilege software with the external debugger. Similarly, trace output from higher privilege execution could leak sensitive information to entities at lower privilege levels if not properly controlled.

Chapter 3. Sdsec (ISA extension)

This chapter introduces the Sdsec ISA extension, which adds security enhancements to the Sdext/Sdtrig [1] and trace functionality [2] [3]. The Sdsec extension is only applicable when external debug (Sdext/Sdtrig) or trace functionality (E-Trace/N-Trace) is implemented in the system. The extension provides mandatory M-mode external debug and trace control, along with optional extensions for lower privilege modes.

When external debug is implemented, the following external debug control extensions are defined:

- **Smmedbgsec (Base)**: Mandatory base extension providing M-mode external debug control
- **Smsedbgsec (Optional)**: Extends Smmedbgsec to add S/HS-mode external debug control
- **Smvsebgsec (Optional)**: Extends Smmedbgsec to add VS-mode external debug control
- **Smuedbgsec (Optional)**: Extends Smmedbgsec to add independent U-mode and VU-mode external debug controls

When trace is implemented, the following trace control extensions are defined:

- **Smmetricsec (Base)**: Mandatory base extension providing M-mode trace control
- **Smsetricsec (Optional)**: Extends Smmetricsec to add S/HS-mode trace control
- **Smvsetrcsec (Optional)**: Extends Smmetricsec to add VS-mode trace control
- **Smuetrcsec (Optional)**: Extends Smmetricsec to add independent U-mode and VU-mode trace controls

The Smmedbgsec base extension is mandatory when external debug is implemented. The Smmetricsec base extension is mandatory when trace is implemented. If a system implements both external debug and trace, both base extensions must be implemented. Implementations may choose to implement some or all of the optional extensions depending on their system requirements. When implementing extensions for lower privilege modes, all extensions for implemented higher privilege modes of the same type (external debug or trace) must also be implemented. These extensions work together to establish hierarchical external debug and trace controls where external debug access or trace output can be restricted to specific privilege levels.



Debug and trace extensions can be implemented independently. For example, a system may implement Smmedbgsec without Smmetricsec, or vice versa.

3.1. External Debug Security Extensions

External debug operations can modify system state and expose sensitive information. The following extensions provide hierarchical privilege-based controls to restrict debug access, preventing unauthorized debugging of higher-privilege software.

3.1.1. External Debug Security Control States and CSRs

The following table summarizes the external debug control states and associated CSRs introduced by the Debug Module Security Extension and the Sdsec external debug security extensions. In this specification, *control state* covers platform-level control state, such as **psecdbgcn**, per-hart platform-controlled states, such as **mdbgcn**, and CSR fields, such as **mdtcfg.SEDBGEN**.

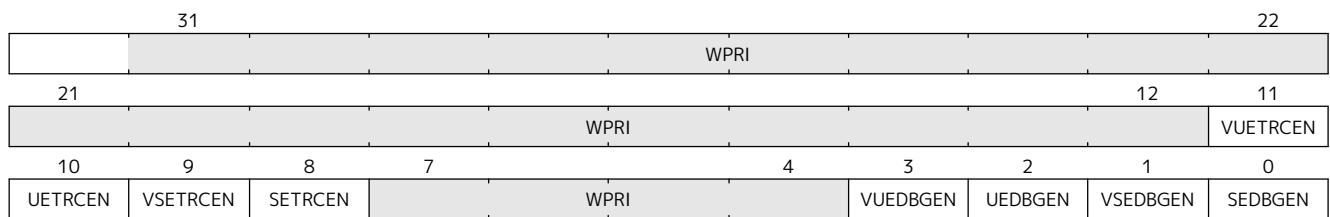
Table 1. External Debug Control States and CSRs

Extension	External Debug Control State	New CSRs	Scope
Debug Module Security Extension	psecdbgcn	None	Platform/all modes

Extension	External Debug Control State	New CSRs	Scope
Smmedbgsec	mdbgcn	mdtcfg	M-mode
Smsedbgsec	mdtcfg, SEDBGEN	sdcsr, sdpc	S/HS-mode
Smvsedbgsec	mdtcfg, VSEDBGEN	None	VS-mode
Smuedbgsec	mdtcfg, UEDBGEN, mdtcfg, VUEDBGEN	udcsr, udpc	U-mode and VU-mode, independently

When external debug is allowed for a privilege mode, it is allowed for all lower privilege modes. U-mode and VU-mode are not ordered with respect to each other, so **UEDBGEN** and **VUEDBGEN** control them independently. If an optional debug extension is not implemented, the corresponding control state behaves as read-only 0; for lower-privilege modes, the **mdtcfg** field representing that control state is also read-only 0.

The **mdtcfg** (Machine Debug and Trace Configuration) is a machine-mode read/write CSR (address TBD) introduced by this specification. It holds configuration fields for external debug and trace security control.



Register 1: **mdtcfg** register

The following fields in **mdtcfg** provide external debug control states for privilege modes other than M-mode. The trace control states are described in [Section 3.2](#).

Table 2. Details of external debug control states in **mdtcfg**

Field	Description	Access	Reset
SEDBGEN	S/HS-mode external debug enable. When set to 1, external debug is allowed for S/HS-mode, U-mode, VS-mode, and VU-mode when mdbgcn is 0. Introduced by Smsedbgsec .	WARL	0
VSEDBGEN	VS-mode external debug enable. When set to 1, external debug is allowed for VS-mode and VU-mode when mdbgcn =0 and SEDBGEN =0. This field does not control U-mode. Introduced by Smvsedbgsec .	WARL	0
UEDBGEN	U-mode external debug enable. When set to 1, U-mode external debug is allowed when mdbgcn =0 and SEDBGEN =0. VSEDBGEN and VUEDBGEN do not affect U-mode permission. Introduced by Smuedbgsec .	WARL	0
VUEDBGEN	VU-mode external debug enable. When set to 1, VU-mode external debug is allowed when mdbgcn =0, SEDBGEN =0, and VSEDBGEN =0. UEDBGEN does not affect VU-mode permission. Introduced by Smuedbgsec .	WARL	0



The bit positions shown above are tentative and have not been officially allocated. The final bit assignments will be determined during the ratification process.

3.1.2. Debug Operations

Chapter 3 of *The RISC-V Debug Specification* [1] outlines all mandatory and optional external debug

operations. The operations listed below are affected by the Sdsec extension; other operations remain unaffected. In the context of this chapter, **debug operations** refer to those listed below.

Debug operations affected by Sdsec:

- Halting the hart to enter Debug Mode
- Executing the Program Buffer
- Serving abstract commands (Access Register, Access Memory)

3.1.3. Debug Access Privilege

The **debug access privilege** is the privilege level used for state accesses performed by the Debug Module or through the Program Buffer, including accesses performed by Abstract Commands and Program Buffer instructions. Register and memory accesses use the privilege level derived in Table 3. Attempts to access state that requires a privilege level above the **debug access privilege** fail and set **abstractcs.CMDERR** to 3.

Table 3. External Debug Configuration and Privilege

mdbgcn	SEDBGEN	VSEDBGEN	UEDBGEN	VUEDBGEN	Debugged Context	Debug Access Privilege
1	Don't care	Don't care	Don't care	Don't care	Any supported mode	M-mode
0	1	Don't care	Don't care	Don't care	S/HS-mode, VS-mode, U-mode, or VU-mode	S/HS-mode
0	0	1	Don't care	Don't care	VS-mode or VU-mode	VS-mode
0	0	Don't care	1	Don't care	U-mode	U-mode
0	0	0	Don't care	1	VU-mode	VU-mode

3.1.4. Allowed Resume Privilege Modes

The privilege modes that can be configured in **dcsr.PRIV** and **dcsr.V** are determined by the table below. If an unsupported or disallowed value is written, the fields retain legal values.

Table 4. Allowed Resume Privilege Modes

Resume Mode	Required Debug Control State
M-mode	mdbgcn =1
S/HS-mode	mdbgcn =1, or mdbgcn =0 and SEDBGEN =1
VS-mode	mdbgcn =1, or mdbgcn =0 and either SEDBGEN =1 or VSEDBGEN =1
U-mode	mdbgcn =1, or mdbgcn =0 and either SEDBGEN =1 or UEDBGEN =1
VU-mode	mdbgcn =1, or mdbgcn =0 and at least one of SEDBGEN , VSEDBGEN , or VUEDBGEN is 1

3.1.5. M-mode External Debug Security Extension (Smmedbgsec)

The Smmedbgsec extension is the mandatory base extension when external debug is implemented.



The **mdbgcn** state is a per-hart platform-controlled state that controls whether M-mode external debug capabilities are available. It may be delivered to the hart through various

methods, such as a new input port, a handshake with the system Root of Trust (RoT), or other methods. The **mdbgen** state for the Root of Trust (RoT) itself should be managed by SoC hardware, likely depending on lifecycle fusing. The implementation can choose to group several harts together and use one signal to drive their **mdbgen** state or assign each hart its own dedicated state. For example, a homogeneous computing system can use a signal to drive all **mdbgen** states to enforce a unified debug policy across all harts.

When **mdbgen** is set to 1, debug is allowed for M-mode and the following rules apply:

- The **debug access privilege** for the hart is M-mode. Abstract Commands, including Quick Access, and instructions in the Program Buffer operate with M-mode privilege.
- The **debug operations** are allowed when the hart executes in any privilege mode.
- Privilege-changing instructions (other than EBREAK) executed in the Program Buffer must raise an exception, stopping execution and setting **abstractcs.CMDERR** to 3.
- The **allowed resume privilege modes** include M-mode and all lower privilege modes.

When **mdbgen** is set to 0 and the hart is running in M-mode, external debug is disallowed for M-mode:

- The hart will not enter Debug Mode:
 - Halt requests will remain pending until external debug is allowed.
 - Triggers with ACTION=1 (enter Debug Mode) will not match or fire.
 - EBREAK cannot enter Debug Mode and always raises a breakpoint exception.
- The external trigger outputs (with ACTION=8/9) will not match or fire.
- Accesses to M-mode accessible CSRs (e.g., **dcsr**, **dpc**, and **dscratch0/1**) will fail and **abstractcs.CMDERR** is set to 3 (exception).
- The configuration in **dcsr** will be ignored as if it were 0.
- DMODE is read/write for accesses from M-mode when Sdtrig is implemented.



*M-mode is given write access to DMODE when **mdbgen** is 0 to allow M-mode software to save and restore trigger context at S/HS-mode software boundaries. This capability is necessary to prevent triggers from serving as a side-channel attack vector against S/HS-mode software where external debugging is disallowed. Without this, an external debugger could indirectly observe states (such as PC and data addresses) from the debug-disallowed software, potentially leading to information leakage or Denial of Service (DoS). By enabling M-mode to switch trigger contexts at privilege mode boundaries, such attacks can be prevented. When **mdbgen** is 1, DMODE remains accessible only from Debug Mode.*

If the hart is running in a debug-allowed lower privilege mode (e.g., S/HS-mode) when **mdbgen** is 0:

- Single-stepping cannot stop in M-mode.
- Interrupts to M-mode cannot be disabled by setting **dcsr.STEPIE**=0.



*When **mdbgen**=0 and **dcsr.STEP**=1, a single-stepped instruction in a debug-allowed privilege mode may transfer control to the M-mode trap handler. The hart will execute the handler in M-mode and re-enter Debug Mode immediately after an MRET instruction returns to the debug-allowed privilege mode (i.e., MRET with **mstatus.MPP**<3). The hart does not re-enter Debug Mode if the MRET instruction returns to a debug-disallowed privilege mode (i.e., MRET with **mstatus.MPP**=3, **mdbgen**=0).*



*This specification assumes that the debugger ensures **mdbgen** shall never be set to 0 while the*

*hart is in Debug Mode. Setting **mdbgen** to 0 while in Debug Mode could lead to undefined behavior; the hart may lose its debug privileges unexpectedly, potentially causing the debug session to fail or become insecure.*

3.1.6. S/HS-mode External Debug Security Extension (Smsedbgsec)

The **Smsedbgsec** extension is optional and requires **Smmedbgsec**. It introduces the **SEDBGEN** field in CSR **mdtcfg** and controls S/HS-mode debuggability when M-mode external debug is disallowed.

The **SEDBGEN** field only takes effect when **mdbgen** is 0.

When **SEDBGEN** is set to 1, S/HS-mode external debug is allowed:

- The **debug access privilege** for the hart is S/HS-mode. Abstract Commands, including Quick Access, and instructions in the Program Buffer operate with S/HS-mode privilege.
- The **debug operations** are allowed when the hart executes in S/HS-mode, U-mode, VS-mode, or VU-mode.
- Privilege-changing instructions (other than EBREAK) executed in the Program Buffer must raise an exception, stopping execution and setting **abstractcs.CMDERR** to 3.
- The **sdcsr** and **sdpc** registers (Section 3.1.6.1) are introduced in **Smsedbgsec** to provide an S/HS-mode debugger with access to **dcsr** and **dpc**.
- The **effective debug access privilege** of loads and stores in Debug Mode may be modified as described in Section 3.1.6.2.
- The **allowed resume privilege modes** include S/HS-mode and all lower privilege modes.

When **SEDBGEN** is set to 0 and the hart is running in S/HS-mode, external debug is disallowed for S/HS-mode:

- The hart will not enter Debug Mode while running in S/HS-mode:
 - Halt requests will remain pending until external debug is allowed.
 - Triggers with ACTION=1 (enter Debug Mode) will not match or fire.
 - EBREAK cannot enter Debug Mode and always raises a breakpoint exception.
- The external trigger outputs (with ACTION=8/9) will not match or fire while in S/HS-mode.
- Accesses to S/HS-mode accessible CSRs (e.g., **sdcsr** and **sdpc**) will fail and **abstractcs.CMDERR** is set to 3 (exception).
- The configuration visible in **sdcsr** will be ignored as if it were 0.

When **SEDBGEN** is set to 0 and the hart is running in a debug-allowed lower privilege mode (e.g., U-mode), the following restrictions apply:

- Single-stepping cannot stop in S/HS-mode.
- Interrupts delegated to S/HS-mode cannot be disabled by setting **dcsr.STEPIE**=0.



*When **SEDBGEN**=0 and **sdcsr.STEP**=1, a single-stepped instruction in a debug-allowed privilege mode may transfer control to the S/HS-mode trap handler. The hart will execute the handler in S/HS-mode and re-enter Debug Mode immediately after an SRET instruction returns to the debug-allowed privilege mode (i.e., SRET with **sstatus.SPP**=0). The hart does not re-enter Debug Mode if the SRET instruction returns to a debug-disallowed privilege mode.*

sdcsr and sdpc

When **SEDBGEN** is 1, the **sdcsr** and **sdpc** registers provide S/HS-mode read/write access to the **dcscr** and **dpc** registers, respectively. However, **sdcsr** does not expose access to the **MPRVEN** field; instead, it repurposes the **MPRVEN** bit position as a **DMPRV** field that modifies the effective debug access privilege in S/HS-mode. Both registers are accessible only in Debug Mode.

Table 5. CSR addresses for S/HS-mode shadows of Debug Mode CSRs

Number	Name	Description
address TBD	sdcsr	S/HS-mode debug control and status register.
address TBD	sdpc	S/HS-mode debug program counter.

The **sdcsr** register exposes a subset of **dcscr**, formatted as shown in [Register 2](#), while the **sdpc** register provides full access to **dpc**.



Unlike **dcscr** and **dpc**, the **dscratch0/1** registers do not have an S/HS-mode access mechanism, and external debuggers with S/HS-mode privilege cannot use them. In The RISC-V Debug Specification [1], **dscratch*** are optional scratch registers that implementations may use internally; **hartinfo.NSCRATCH** indicates how many are available, and a debugger must not write to these registers unless **hartinfo** explicitly mentions them. Compliant debug flows already work without **dscratch** (for example, Access Register, GPRs, and data registers). Sdsec shadows only the required control state at each privilege level (**dcscr/dpc** → **sdcsr/sdpc** and, when applicable, **udcsr/udpc**); no **sdscratch** registers are defined. In an execution-based implementation, the park loop runs at M-mode and may use existing **dscratch** registers, while Abstract Commands and the Program Buffer run at the [debug access privilege](#).

31	28	27	26	24	23	22
DEBUGVER			0	EXTCAUSE		0
21	19	18	17	16	15	14
0	PELP	EBREAKVS	EBREAKVU	0	EBREAKS	EBREAKU
10	9	8	6	5	4	3
0	CAUSE		V	DMPRV	0	STEP
					0	PRV

Register 2: S/HS-mode debug control and status register (sdcsr)



The **NMIP**, **MPRVEN**, **STOPTIME**, **STOPCOUNT**, **EBREAKM**, and **CETRIG** fields in **dcscr** are configurable only by M-mode; they are masked in **sdcsr**, while **PRV[1]** is hardwired to 0 in **sdcsr**. The field for **MPRVEN** is reclaimed by **DMPRV** in the **sdcsr** layout to avoid wasting fields.

Table 6. Details of the **dmprv** field in **sdcsr**

Field	Description	Access	Reset
DMPRV	0 (normal): The privilege level in Debug Mode is not modified. 1: In Debug Mode, the privilege level for load and store operations is modified to the effective debug access privilege as described in Section 3.1.6.2 and Section 3.1.7.2 .	WARL	0

Debug Access Privilege for Memory in Smsedbgsec

The **sdcsr.DMPRV** field takes effect when **mdbgen** is 0 and is read-only 0 when **mdbgen** is 1. When **SEDBGEN** is 1, **sdcsr.DMPRV** can modify the effective debug access privilege used by an S/HS-mode debugger's loads and stores in Debug Mode. When **sdcsr.DMPRV**=0, the effective debug access privilege of loads and

stores in Debug Mode follows [Table 3](#); when **sdcscr.DMPRV**=1, the **effective debug access privilege** of loads and stores in Debug Mode is represented by **sstatus.SPP** and **hstatus.SPV** (if the hypervisor extension is supported).

The **sdcscr.DMPRV** does not affect the virtual-machine load/store instructions, HLV, HLVX, and HSV.

3.1.7. VS-mode External Debug Security Extension (Smvsedbgsec)

The Smvsedbgsec extension is optional and requires Smsedbgsec. It introduces the **VSEDBGGEN** field in CSR **mdtcfg** and controls VS-mode debuggability when S/HS-mode external debug is disallowed.

The **VSEDBGGEN** field only takes effect when both **mdbggen** is 0 and **SEDBGGEN** is 0.

When **VSEDBGGEN** is set to 1, VS-mode external debug is allowed:

- The **debug access privilege** for the hart is VS-mode. Abstract Commands, including Quick Access, and instructions in the Program Buffer operate with VS-mode privilege.
- The **debug operations** are allowed when the hart executes in VS-mode or VU-mode.
- Privilege-changing instructions (other than EBREAK) executed in the Program Buffer must raise an exception, stopping execution and setting **abstractcs.CMDERR** to 3.
- The **sdcscr** and **sdpc** registers (from Smsedbgsec) provide VS-mode read/write access to **dscr** and **dpc**, respectively, as specified in [Section 3.1.7.1](#).
- The **effective debug access privilege** of loads and stores in Debug Mode may be modified as described in [Section 3.1.7.2](#).
- The **allowed resume privilege modes** include VS-mode and VU-mode.

When **VSEDBGGEN** is set to 0 and the hart is running in VS-mode, external debug is disallowed for VS-mode:

- The hart will not enter Debug Mode while running in VS-mode:
 - Halt requests will remain pending until external debug is allowed.
 - Triggers with ACTION=1 (enter Debug Mode) will not match or fire.
 - EBREAK cannot enter Debug Mode and always raises a breakpoint exception.
- The external trigger outputs (with ACTION=8/9) will not match or fire while in VS-mode.
- Accesses to VS-mode accessible CSRs (e.g., **sdcscr** and **sdpc**) will fail and **abstractcs.CMDERR** is set to 3 (exception).
- The configuration visible to VS-mode through **sdcscr** will be ignored as if it were 0.

When **VSEDBGGEN** is set to 0 and the hart is running in a debug-allowed VU-mode, the following restrictions apply:

- Single-stepping cannot stop in VS-mode.
- Interrupts delegated to VS-mode cannot be disabled by setting **sdcscr.STEPIE**=0.



*When **VSEDBGGEN**=0 and **sdcscr.STEP**=1, a single-stepped instruction in a debug-allowed VU-mode may transfer control to the VS-mode trap handler. The hart will execute the handler in VS-mode and re-enter Debug Mode immediately after an SRET instruction returns to the debug-allowed VU-mode. The hart does not re-enter Debug Mode if the SRET instruction returns to a debug-disallowed VS-mode.*

sdcsr and sdpc in VS-mode

When **VSEDBGEN** is 1, the **sdcsr** and **sdpc** (Section 3.1.6.1) are accessible with VS-mode privilege, providing access to the **dcscr**. The **sdcsr.EBREAKS** and **sdcsr.EBREAKU** fields are redirected to **dcscr.EBREAKVS** and **dcscr.EBREAKVU**, while writes to **sdcsr.EBREAKVS**, **sdcsr.EBREAKVU**, and **sdcsr.V** are discarded (writes are ignored and reads return 0). Similar to **sdcsr** access when **SEDBGEN** is 1, **sdcsr.DMPRV** modifies the effective debug access privilege in VS-mode.



*Redirected access to **dcscr.EBREAKVS** and **dcscr.EBREAKVU** unifies the configuration for both S/HS-mode and VS-mode. However, the virtualization mode cannot be changed through **sdcsr.V** by a VS-mode debugger.*

Debug Access Privilege for Memory in Smvsedbgsec

The **sdcsr.DMPRV** modifies the effective debug access privilege of loads and stores for a VS-mode debugger when **SEDBGEN** is 0 and **VSEDBGEN** is 1.

When **sdcsr.DMPRV**=0, the effective debug access privilege of loads and stores in Debug Mode follows Table 3; when **sdcsr.DMPRV**=1, the effective debug access privilege of loads and stores in Debug Mode is represented by **vsstatus.SPP** with the virtualization mode being honored as 1.

3.1.8. U-mode and VU-mode External Debug Security Extension (Smuedbgsec)

The **Smuedbgsec** extension is optional. It requires **Smmedbgsec** and, on harts that implement S/HS-mode, **Smsedbgsec**. When VU-mode control is implemented, **Smuedbgsec** also requires **Smvsedbgsec**. **Smvsedbgsec** is not required when only U-mode control is implemented and **VUEDBGEN** is read-only 0. **Smuedbgsec** introduces independent **UEDBGEN** and **VUEDBGEN** fields in CSR **mdtcfg**. If a mode or its corresponding control is not implemented, the corresponding field is read-only 0.

U-mode External Debug Control

The **UEDBGEN** field takes effect when **mdbggen**=0 and **SEDBGEN**=0. Because U-mode and VS/VU-mode are not ordered with respect to each other, **VSEDBGEN** and **VUEDBGEN** do not affect U-mode external debug permission.

When **UEDBGEN**=1, **mdbggen**=0, and **SEDBGEN**=0, U-mode external debug is allowed:

- The **debug access privilege** is U-mode. Abstract Commands and instructions in the Program Buffer operate with U-mode privilege.
- The **debug operations** are allowed when the hart executes in U-mode.
- Privilege-changing instructions (other than **EBREAK**) executed in the Program Buffer must raise an exception, stopping execution and setting **abstractcs.CMDERR** to 3.
- The **udcsr** and **udpcc** registers described in Section 3.1.8.3 provide access to the permitted **dcscr** and **dpc** state.
- U-mode is the only **allowed resume privilege mode**.

When **UEDBGEN** is set to 0 and the hart is running in U-mode, external debug is disallowed for U-mode:

- The hart will not enter Debug Mode while running in U-mode:
 - Halt requests remain pending until external debug is allowed.
 - Triggers with **ACTION**=1 do not match or fire.

- EBREAK raises a breakpoint exception instead of entering Debug Mode.
- External trigger outputs with ACTION=8 or ACTION=9 do not match or fire.
- Accesses to user-visible debug CSRs fail and set **abstractcs.CMDERR** to 3.
- The configuration visible through **udcsr** is ignored as if it were 0.

VU-mode External Debug Control

The **VUEDBGEN** field takes effect when **mdbgen**=0, **SEDBGEN**=0, and **VSEDBGEN**=0. **UEDBGEN** does not affect VU-mode external debug permission.

When **VUEDBGEN**=1, **mdbgen**=0, **SEDBGEN**=0, and **VSEDBGEN**=0, VU-mode external debug is allowed:

- The [debug access privilege](#) is VU-mode. Abstract Commands and instructions in the Program Buffer operate with VU-mode privilege.
- The [debug operations](#) are allowed when the hart executes in VU-mode.
- Privilege-changing instructions (other than EBREAK) executed in the Program Buffer must raise an exception, stopping execution and setting **abstractcs.CMDERR** to 3.
- The **udcsr** and **udpc** registers described in [Section 3.1.8.3](#) provide access to the permitted **dcsr** and **dpc** state.
- VU-mode is the only [allowed resume privilege mode](#).

When **VUEDBGEN** is set to 0 and the hart is running in VU-mode, external debug is disallowed for VU-mode:

- The hart will not enter Debug Mode while running in VU-mode:
 - Halt requests remain pending until external debug is allowed.
 - Triggers with ACTION=1 do not match or fire.
 - EBREAK raises a breakpoint exception instead of entering Debug Mode.
- External trigger outputs with ACTION=8 or ACTION=9 do not match or fire.
- Accesses to user-visible debug CSRs fail and set **abstractcs.CMDERR** to 3.
- The configuration visible through **udcsr** is ignored as if it were 0.

udcsr and udpc

The **udcsr** and **udpc** registers provide U-mode or VU-mode read/write access to the **dcsr** and **dpc** state when the corresponding **UEDBGEN** or **VUEDBGEN** control allows debug. The **udcsr** exposes the subset of **dcsr** shown in [Register 3](#). Access to **udcsr.EBREAKU** is redirected to **dcsr.EBREAKVU** when the virtualization mode is 1. The **udpc** register provides full access to **dpc**.



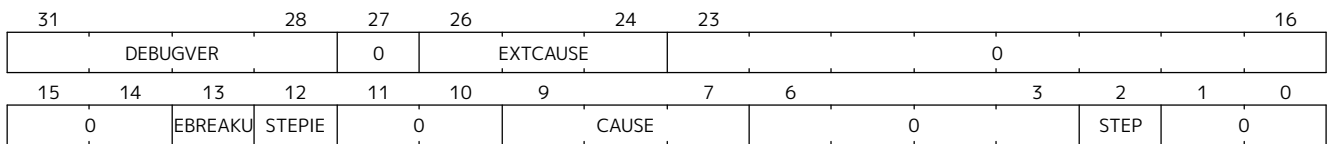
*The **udcsr** does not expose **dcsr.V**. A U-mode or VU-mode debugger therefore cannot switch between U-mode and VU-mode.*

Table 7. CSR addresses for U/VU-mode shadows of Debug Mode CSRs

Number	Name	Description
0xaaa	udcsr	U-mode or VU-mode debug control and status register.
0xaaa	udpc	U-mode or VU-mode debug program counter.

The **udcsr** register exposes a subset of **dcsr**, formatted as shown in [Register 3](#), while the **udpc** register

provides full access to **dpc**.



Register 3: U-mode or VU-mode debug control and status register (*udcsr*)

3.2. Trace Security Extensions

When trace is implemented, trace output can expose sensitive information across privilege boundaries. The extensions below provide security controls to restrict trace output based on privilege levels.

3.2.1. Trace Security Control States

The following table defines the trace control states introduced by the trace security extensions. In this specification, *control state* covers both platform-controlled states, such as **mtrcen**, and CSR fields, such as **mdtcfg.SETRCEN**.

Table 8. Trace Control States

Extension	Trace Control State	Target Modes
Smmetrcsec	mtrcen	M-mode
Smsetrcsec	mdtcfg.SETRCEN	S/HS-mode
Smvsetrcsec	mdtcfg.VSETRCEN	VS-mode
Smuetrosec	mdtcfg.UETRCEN, mdtcfg.VUETRCEN	U-mode and VU-mode, independently

Allowing trace for a privilege mode also allows it for all lower privilege modes. U-mode and VU-mode are not ordered with respect to each other, so **UETRCEN** and **VUETRCEN** control them independently. Each **mdtcfg** control field introduced by an unimplemented optional trace extension is read-only 0.



The availability of trace output is controlled through signals defined in the RISC-V hart-trace interface (RHTI) [4]. An implementation can convert the logical **sec_inhibit** signal to the canonical trace interface signals.

The following **mdtcfg** fields provide trace control states for privilege modes other than M-mode. The **mdtcfg** CSR must be implemented when **Smmetrcsec** is implemented, even if **Smmedbgsec** is not implemented.

Table 9. Details of trace security control states in **mdtcfg**

Field	Description	Access	Reset
SETRCEN	S/HS-mode external trace enable. When set to 1, trace is allowed for S/HS-mode, U-mode, VS-mode, and VU-mode when mtrcen is 0. Introduced by Smsetrcsec .	WARL	0
VSETRCEN	VS-mode external trace enable. When set to 1, trace is allowed for VS-mode and VU-mode when mtrcen =0 and SETRCEN =0. This field does not control U-mode. Introduced by Smvsetrcsec .	WARL	0
UETRCEN	U-mode external trace enable. When set to 1, U-mode trace is allowed when mtrcen =0 and SETRCEN =0. VSETRCEN and VUETRCEN do not affect U-mode permission. Introduced by Smuetrosec .	WARL	0

Field	Description	Access	Reset
VUETRCEN	VU-mode external trace enable. When set to 1, VU-mode trace is allowed when mtrcen =0, SETRCEN =0, and VSETRCEN =0. UETRCEN does not affect VU-mode permission. Introduced by Smuetrcsec.	WARL	0

3.2.2. M-mode Trace Security Extension (Smmetricsec)

The Smmetricsec extension is the mandatory base extension when trace is implemented.



*The **mtrcen** state is a per-hart platform-controlled state that controls whether M-mode trace output is enabled. It may be delivered to the hart through various methods, such as a new input port, a handshake with the system Root of Trust (RoT), or other methods. The **mtrcen** state for the Root-of-Trust (RoT) itself should be managed by SoC hardware, likely dependent on lifecycle fusing. The implementation can choose to group several harts together and use one signal to drive their **mtrcen** state or assign each hart its own dedicated state. For example, a homogeneous computing system can use a signal to drive all **mtrcen** states to enforce a unified trace policy across all harts.*

When **mtrcen** is set to 1, trace is allowed for all privilege modes. When **mtrcen** is set to 0, trace is inhibited for M-mode by asserting the logical **sec_inhibit** signal to the trace encoder when the hart is executing in M-mode.

3.2.3. S/HS-mode Trace Security Extension (Smsetrcsec)

The Smsetrcsec extension is optional and requires Smmetricsec. It introduces the **SETRCEN** field in CSR **mdtcf**g and controls S/HS-mode trace output when M-mode trace is disallowed.

The **SETRCEN** field only takes effect when **mtrcen** is 0.

When **SETRCEN** is set to 1, trace is allowed for all privilege modes except M-mode. When **SETRCEN** is set to 0, trace is inhibited for S/HS-mode by asserting the logical **sec_inhibit** signal to the trace encoder when the hart is executing in S/HS-mode.

3.2.4. VS-mode Trace Security Extension (Smvsetrcsec)

The Smvsetrcsec extension is optional and requires Smsetrcsec. It introduces the **VSETRCEN** field in CSR **mdtcf**g and controls VS-mode trace output when S/HS-mode trace is disallowed.

The **VSETRCEN** field only takes effect when both **mtrcen** is 0 and **SETRCEN** is 0.

When **VSETRCEN** is set to 1, trace is allowed for VS-mode and VU-mode. When **VSETRCEN** is set to 0, trace is inhibited for VS-mode by asserting the logical **sec_inhibit** signal to the trace encoder when the hart is executing in VS-mode.

3.2.5. U-mode and VU-mode Trace Security Extension (Smuetrcsec)

The Smuetrcsec extension is optional. It requires Smmetricsec and, on harts that implement S/HS-mode, Smsetrcsec. When VU-mode trace control is implemented, Smuetrcsec also requires Smvsetrcsec. Smvsetrcsec is not required when only U-mode trace control is implemented and **VUETRCEN** is read-only 0. Smuetrcsec introduces independent **UETRCEN** and **VUETRCEN** fields in CSR **mdtcf**g. If a mode or its corresponding trace control is not implemented, the corresponding field is read-only 0.

The **UETRCEN** field takes effect when **mtrcen**=0 and **SETRCEN**=0. **VSETRCEN** and **VUETRCEN** do not affect U-mode trace permission. When **UETRCEN** is 1, trace is allowed in U-mode; when it is 0, trace is inhibited in U-mode by asserting the logical **sec_inhibit** signal.

The **VUETRCEN** field takes effect when **mtrcen**=0, **SETRCEN**=0, and **VSETRCEN**=0. **UETRCEN** does not affect VU-mode trace permission. When **VUETRCEN** is 1, trace is allowed in VU-mode; when it is 0, trace is inhibited in VU-mode by asserting the logical **sec_inhibit** signal.

Chapter 4. Debug Module Security (non-ISA) Extension

This chapter defines the required security enhancements for the Debug Module. The debug operations listed below are modified by this extension.

- Halt
- Reset
- Keepalive
- Abstract commands (Access Register, Quick Access, Access Memory)
- System bus access

If any hart in the system implements the Sdsec extension, the Debug Module must also implement the Debug Module Security Extension.

4.1. External Debug Security Extensions Discovery

The ISA and non-ISA external debug security extensions impose security constraints and introduce non-backward-compatible changes. The presence of the extensions can be determined by polling the ALLSECURED and/or ANYSECURED bits in **dmstatus** Table 10. These bits will read as 0 when **psecdbgen** is 0, regardless of whether the harts implement the Sdsec extension. When **psecdbgen** is 1, ALLSECURED is set to 1 if all currently selected harts implement the Sdsec extension, and ANYSECURED is set to 1 if any currently selected hart implements the Sdsec extension. When any hart implements the Sdsec extension, it indicates that the Debug Module implements the Debug Module Security Extension as described in this chapter.

4.2. Halt

The halt behavior for a hart is detailed in Section 3.1. According to *The RISC-V Debug Specification* [1], a halt request must be responded to within one second. However, this constraint must be relaxed, as the request might remain pending when debugging is disallowed. In the case of a halt-on-reset request (SETRESETHALTREQ), the request is only acknowledged by the hart once it has reached a privilege level in which debug is allowed.

4.3. Reset

The HARTRESET operation resets selected harts. When M-mode debugging is disallowed, the hart will raise a security fault error to the Debug Module on HARTRESET operations. The error can be detected through ALLSECFAULT and ANYSECFAULT in **dmstatus**.

The NDMRESET operation is a system-level reset not tied to hart privilege levels and resets the entire system (excluding the Debug Module). The Debug Module Security Extension makes the NDMRESET field read-only 0 when **psecdbgen** is 1. The debugger can determine support for the NDMRESET operation by setting the field to 1 and subsequently verifying the returned value upon reading.

4.4. Keepalive

The SETKEEPALIVE bit serves as an optional request for the hart to remain available for debugging. This bit only takes effect when M-mode debugging is allowed; otherwise, the hart behaves as if the bit is not set.



Keepalive affects whole-hart power and availability, including M-mode. Halt and single-step

requests with `mdbgen=0` remain privilege-scoped and do not imply hart-global power control. Therefore, a lower-privilege debug enable does not grant the debugger authority to keep the hart available through `keepalive`, and `SETKEEPALIVE` remains read-only 0 when `mdbgen=0`. If `keepalive` is needed later, it should be introduced as separate M-mode delegation extension rather than implied by `mdtcfg.*EN`.

4.5. Abstract Commands

The hart's response to abstract commands is detailed in [Section 3.1](#). The following subsection delineates the constraints when the Debug Module issues an abstract command.

4.5.1. Relaxed Permission Check `reLaxedpriv`

The `reLaxedpriv` field is hardwired to 0.

4.5.2. Address Translation `AAMVIRTUAL`

The field `command.AAMVIRTUAL` determines whether the Access Memory command uses a physical or virtual address. When `mdbgen=1`, the behavior follows the original RISC-V Debug Specification [1]. When `mdbgen=0`:

- If `AAMVIRTUAL=0`, the hart responds with a security fault error (setting `abstractcs.CMDERR` to 6).
- If `AAMVIRTUAL=1`, addresses are treated as virtual and are translated and protected according to the debug access privilege, as defined in [Section 3.1.3](#), and may be modified to the effective debug access privilege as defined in [Section 3.1.6.2](#) or [Section 3.1.7.2](#).

4.5.3. Quick Access

When M-mode debugging is disallowed (`mdbgen=0`) for a hart, any Quick Access operation will be discarded by the Debug Module, causing `abstractcs.CMDERR` to be set to 6.



Quick Access abstract commands initiate a halt, execute the Program Buffer, and resume the selected hart. These halts cannot remain pending until debugging is allowed because the debugger blocks while waiting for completion. To address this, the hart would need to distinguish between Quick Access and other halt requests. Since Quick Access is an optional optimization, the Debug Module Security Extension avoids this additional hardware complexity by disallowing Quick Access when `mdbgen` is 0.

4.6. System Bus Access

The System Bus Access must be checked by bus initiator protection mechanisms such as IOPMP [5] or RISC-V Worlds [6]. The bus protection unit can return an error to the Debug Module on illegal accesses; in that case, the Debug Module will set `sbc.SBERROR` to 6 (security fault error).



Trusted entities like the RoT should configure IOPMP or equivalent protection before external debug is allowed.

4.7. Security Fault Error Reporting

A dedicated error code, security fault error (CMDERR 6), is included in `abstractcs.CMDERR`. Issuance of

abstract commands under disallowed circumstances sets **abstractcs.CMDERR** to 6. Additionally, the bus security fault error (SBERROR 6) is introduced in **sbcs.SBERROR** to denote errors related to system bus access.

Error status bits are internally maintained for each hart. The ALLSECFAULT and ANYSECFAULT fields in **dmstatus** indicate the error status of the currently selected harts. These error statuses are sticky and can only be cleared by writing 1 to **dmcs2.ACKSECFAULT** [Table 11](#).

4.8. Platform Debug Security Enable

The platform state element **psecdbgen** is introduced to enable the debug security constraints defined by this specification. When **psecdbgen** is 0, the platform retains full debugging capabilities, as if the extensions in this specification were not implemented:

- All [debug operations](#) are executed with M-mode privilege (equivalent to having **mdbgen** set to 1) for all harts in the system.
- The NDMRESET operation is allowed.
- The RELAXEDPRIV field may be configurable.
- System Bus Access may bypass the bus initiator protections.
- Trace output is enabled in all privilege modes.

When **psecdbgen** is 1, the security constraints in this specification apply.

4.9. Update of Debug Module Registers

The Debug Module Security Extension introduces new fields in the Debug Module registers. In **dmstatus**, ANYSECURED and ALLSECURED are added at bit 20 and bit 21 respectively, while ANYSECFAULT and ALLSECFAULT are added at bit 25 and bit 26. The ACKSECFAULT field is added to **dmcs2** at bit 12.

Table 10. Details of newly introduced fields in **dmstatus**

Field	Description	Access	Reset
ALLSECURED	The field is 0 when psecdbgen is 0. When psecdbgen is 1, this field is 1 if all currently selected harts implement the Sdsec extension.	R	-
ANYSECURED	The field is 0 when psecdbgen is 0. When psecdbgen is 1, this field is 1 if any currently selected hart implements the Sdsec extension.	R	-
ALLSECFAULT	The field is 1 when all currently selected harts have raised a security fault due to reset operation	R	-
ANYSECFAULT	The field is 1 when any currently selected hart has raised a security fault due to reset operation	R	-

Table 11. Detail of ACKSECFAULT in **dmcs2**

Field	Description	Access	Reset
ACKSECFAULT	0 (nop): No effect. 1 (ack): Clears error status bits for any selected harts.	W1	-

Appendix A: Valid Extension Combinations

This appendix lists valid external debug and trace extension combinations for the supported privilege-mode architectures.

A.1. External Debug Security Extension Combinations

Table [Table 12](#) shows valid external debug extension combinations for different architectures:

Table 12. Valid External Debug Extension Combinations

Architecture	Valid Debug Extension Combinations
M-only	• Smmedbgsec
M/U	• Smmedbgsec • Smmedbgsec + Smuedbgsec
M/S/U	• Smmedbgsec • Smmedbgsec + Smsedbgsec • Smmedbgsec + Smsedbgsec + Smuedbgsec
M/S/VS/VU/U	• Smmedbgsec • Smmedbgsec + Smsedbgsec • Smmedbgsec + Smsedbgsec + Smuedbgsec • Smmedbgsec + Smsedbgsec + Smvsgedbgsec • Smmedbgsec + Smsedbgsec + Smvsgedbgsec + Smuedbgsec

A.2. Trace Security Extension Combinations

Table [Table 13](#) shows valid trace extension combinations for different architectures:

Table 13. Valid Trace Extension Combinations

Architecture	Valid Trace Extension Combinations
M-only	• Smmetricsec
M/U	• Smmetricsec • Smmetricsec + Smuetricsec
M/S/U	• Smmetricsec • Smmetricsec + Smsetrcsec • Smmetricsec + Smsetrcsec + Smuetricsec
M/S/VS/VU/U	• Smmetricsec • Smmetricsec + Smsetrcsec • Smmetricsec + Smsetrcsec + Smuetricsec • Smmetricsec + Smsetrcsec + Smvsetrcsec • Smmetricsec + Smsetrcsec + Smvsetrcsec + Smuetricsec

Appendix B: Software Debug/Trace Security Policy Management

This section provides guidance on how software at different privilege levels can manage debug/trace security policies.

B.1. M-mode Management

M-mode software is responsible for managing the **SEDBGEN**, **VSEDBGEN**, **UEDBGEN**, and **VUEDBGEN** fields in the **mdtcfg** CSR for external debug, and the **SETRCEN**, **VSETRCEN**, **UETRCEN**, and **VUETRCEN** fields in the **mdtcfg** CSR for trace. The **mdbggen** and **mtrcen** states are typically controlled by platform hardware, such as an RoT or lifecycle fuses, and are read-only from a software perspective.

M-mode software should:

1. Initialize debug/trace security policy for lower privilege modes based on security requirements.
2. Provide interfaces for dynamic debug/trace security policy updates (e.g., through SBI calls from lower privilege modes).
3. Save and restore debug/trace security policies when switching between supervisor software contexts.
4. Save and restore trigger context (using DMODE access when **mdbggen**=0) during supervisor software context switches to prevent side-channel attacks.

B.2. S/HS-mode Management

S/HS-mode software is responsible for managing the debug/trace security policy for lower privilege modes:

1. Call M-mode to apply the debug/trace security policy before launching an application or VM.
2. Provide interfaces to handle dynamic debug/trace security policy update requests (e.g., SBI calls from VMs) and call M-mode to apply the security policy accordingly.
3. Save and restore debug/trace security policies when switching between applications or VMs.

B.3. VS-mode Management

VS-mode software is responsible for managing the debug/trace security policy for applications in VU-mode:

1. Call S/HS-mode to apply the debug/trace security policy before launching an application.
2. Save and restore debug/trace security policies when switching between applications.

Appendix C: Execution-Based Implementation with Sdsec

In an execution-based implementation, the code executing the "park loop" can always run with M-mode privilege to access memory and CSRs. However, once execution is dispatched to an Abstract Command or the Program Buffer, the privilege level used to access memory and CSRs should be restricted to the [debug access privilege](#).

To achieve this, a Debug-Mode-only state element (e.g., a field in a custom CSR) may be introduced to control the privilege level in Debug Mode. When the state is set to 1, Debug Mode allows M-mode privilege; when cleared to 0, it enforces the [debug access privilege](#). The hardware sets this state to 1 upon entering the park loop and clears it to 0 with the final instruction of the park loop, immediately before execution is transferred to an Abstract Command or the Program Buffer.

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